Kalsang Lama

Cybersecurity Student & Developer

Kathmandu, Nepal | kalsanglama200@gmail.com | kalsanglama.com.np
github.com/kalsanglama200 | linkedin.com/in/kalsang-lama-894715310

PROFILE

Cybersecurity student who builds and tests security tooling: a multi-tenant PKI and digital-signature platform, and a static malware-analysis engine mapped to MITRE ATT&CK. Strengths in secure application development, access control, vulnerability assessment, digital forensics and Linux/network security. Seeking a cybersecurity role or internship working on secure public-sector digital services.

EDUCATION

BSc (Hons) Ethical Hacking and Cyber Security — Expected Graduation: 2027 2024 – 2027
Softwarica College of IT and E-Commerce, Kathmandu | Practical Pen-Testing, Digital Forensics, Computer Systems & Networks

+2 Management with Computer Science 2021 – 2023
Arunima College | GPA 3.44 / 4.00

PROJECTS

SecureSign Enterprise — PKI & Digital Signature Platform

Source code: github.com/kalsanglama200/secure-signin

- **Implemented** an internal Certificate Authority that issues and revokes X.509 certificates; **built** RSA-PSS/SHA-256 document signing with verification of signature, SHA-256 integrity and certificate revocation status.
- **Secured** access with JWT (issuer, audience, expiry and signature checks), Argon2 password hashing, admin/user RBAC and per-organisation tenant isolation; cross-tenant requests return not-found to avoid resource disclosure.
- **Hardened** the Docker/Nginx deployment: TLS 1.2/1.3 with HTTP-to-HTTPS redirect, 6 security headers (HSTS, CSP, X-Frame-Options and others), login rate limit of 5 requests/min, upload validation (extension, size, content) and audit logging of 7 security event types.
- **Tested** 24 documented scenarios, including cross-tenant access/download/signing, RBAC, rate limiting and an end-to-end upload, sign, verify and audit flow. Documented known limits (no HSM, JWT in session storage). Addresses document tampering, unauthorised access, repudiation and brute-force login.

  Tech: Python, FastAPI, React, PostgreSQL, Docker, Nginx, SQLAlchemy, Alembic, PyJWT, Cryptography

Advanced Malware Detection & Threat Intelligence System

Source code: github.com/kalsanglama200/Malware-detection-and-analysis

- **Developed** a Python static-analysis scanner combining SHA-256 hash matching, entropy/base64-obfuscation heuristics, packed-binary detection, IOC extraction (URLs, IPs, suspicious commands) and optional VirusTotal reputation lookups.
- **Implemented** weighted risk scoring (0–100) with configurable suspicious/malicious thresholds, confidence rating, MITRE ATT&CK technique mapping, file quarantine and JSON reports; **added** an analyst false-positive feedback loop and a GUI with 9 report views.
- **Validated** with a 9-function automated test suite (~1,500 lines of Python). Limitation: static analysis only; no sandboxed execution.

  Tech: Python, CustomTkinter, VirusTotal API, MITRE ATT&CK, entropy analysis, hashing

Freelancing Platform (coursework) — Full-stack web application

Source code: github.com/kalsanglama200/Web_Coursework

- **Built** a React/Node.js platform with user authentication, project posting and interactive features.

TECHNICAL SKILLS

Security Assessment	Penetration testing (web exploitation), vulnerability assessment, reverse engineering (Ghidra), network analysis (Wireshark)
Incident Response	Digital forensics and log analysis (Autopsy, FTK, Volatility), incident reconstruction following NIST SP 800-86
Network Security	Segmentation, firewalls, VPN, access control, intrusion prevention; Cisco Packet Tracer
Secure Development	Python, FastAPI, React, SQL; JWT, RBAC, PKI/X.509, input validation, secure coding
Linux & Infrastructure	Linux administration, Docker, Nginx, PostgreSQL, Bash

EXPERIENCE

Computer Science Teacher 2023 – 2024
Progressive School, Mahankal

- **Taught** computer science and programming to Grade 9–10 students.

CERTIFICATIONS & COMPETITIONS

- **Certified Cybersecurity Educator Professional**, Red Team Leaders, 2025 — verify
- **Pre Security**, TryHackMe, 2025 — verify
- **Capture the Flag:** placed 2nd and 3rd in cybersecurity CTF competitions.